



جامعة الحسين التقنية
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Designing and Implementing Machine Learning Model

Technical Report

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1 Data Preparation and Description (Q1)

1.1 Data Description

- The data in the Gallstone dataset was obtained from the International Medicine Outpatient Clinic of Ankara VM Medical Park Hospital during the time period of June 2022 to June 2023. The Gallstone dataset is made up of 319 patients where 161 of these have been identified as having gallstone disease (classified as "0" for gallstone disease) and 158 are classified as being free of gallstones ("1") or controls. There are also 38 different features included in the Gallstone dataset, which can be categorized into demographics, bio-impedance and lab tests.

Data Information:

- Patients: Total – 319.
- **Present with Gallstones ("0"):** 161 (50.5%).
- **Absent with Gallstones ("1"):** 158 (49.5%).
- **Features:** 38.
- **Target Feature:** Gallstone Status “Categorical, Binary [0 has gallstones, 1 no gallstones]”.
- Nearly balanced data classes.

The 38 features can be presented as 4 categories:

- **Demographical:** Age, Gender, Height, Weight, BMI.
- **Bioimpedance:** Total Body Water, Extracellular Water, Intracellular Water, Lean Mass, Body Fat Mass, Protein, Visceral Fat Area, Hepatic Fat Accumulation.
- **Laboratory Test Results:** Glucose, Total Cholesterol, High-Density Lipoprotein (HDL), Low-Density Lipoprotein (LDL), Triglycerides, Aspartate Aminotransferase (AST), Alanine Aminotransferase (ALT), Alkaline Phosphatase (ALP), Creatinine, Glomerular Filtration Rate (GFR), C-Reactive Protein (CRP), Haemoglobin, VitaminD.
- **Medical History:** Comorbidity status, Coronary Artery Disease (CAD), Hypothyroidism, Hyperlipidaemia, Diabetes Mellitus (DM), Obesity, and other clinical conditions.

Problem Statement:

Gallstones are a common cause of hepatobiliary disorders affecting millions of individuals world-wide with potential serious consequences. Although traditional diagnostic techniques, such as ultrasonography, CT, and MRI, detect gallstones, they have some limitations, including high cost and potential inaccuracies in certain populations. We should propose a technique that offers a viable alternative to conventional imaging techniques for early prediction of gallstone disease.

The overall purpose of this project is to develop a machine learning-based predictive model capable of identifying patients who will be at increased risk of developing gallstones using non-invasive clinical measurements. The specific goal of this study

was to create a binary classifier which identifies a patient as having gallstones or not based upon the patient's demographic data, bioimpedance measurements and laboratory test results (Irfan Esen, 2024).

1.2 Data Preparation

Data preparation is the process that contains Preprocessing and Cleaning. It is the process of cleaning and organizing the raw data so it can be used by a machine-learning models. Effective data preprocessing improves model accuracy, removes noise and inconsistent data, handles missing values, ensure data compatibility, prevent data leakage, and finally it reduces computational costs and shortens development cycles (Mitchell, 2025).

These are the steps I followed in data preparation:

- **Step 1: Data Loading**

The dataset was loaded from UCI Machine Learning Repository. After that, the first examination of the dataset revealed 319 samples of data, a total of 38 input features, and a single output feature. A thorough understanding of the data will prevent an incorrect assumption about the data and ensure the proper techniques are applied to preprocess the data.

```
df=pd.read_csv(r'C:\Users\aalma\Downloads\dataset-ucicSV.csv')
df.head()
```

✓ 0.0s

	Gallstone Status	Age	Gender	Comorbidity	Coronary Artery Disease (CAD)	Hypothyroidism	Hyperlipidemia	Diabetes Mellitus (DM)	Height	Weight	...	High Density Lipoprotein (HDL)	Triglyceride	Aspartat Aminotransferaz (AST)	Alanin Aminotransferaz (ALT)	Alkaline Phosphatase (ALP)	Cres
0	0	50	0	0	0	0	0	0	185	92.8	...	40.0	134.0	20.0	22.0	87.0	
1	0	47	0	1	0	0	0	0	176	94.5	...	43.0	103.0	14.0	13.0	46.0	
2	0	61	0	0	0	0	0	0	171	91.1	...	43.0	69.0	18.0	14.0	66.0	
3	0	41	0	0	0	0	0	0	168	67.7	...	59.0	53.0	20.0	12.0	34.0	
4	0	42	0	0	0	0	0	0	178	89.6	...	30.0	326.0	27.0	54.0	71.0	

5 rows x 39 columns

```
df.columns
```

✓ 0.0s

Index(['Gallstone Status', 'Age', 'Gender', 'Comorbidity', 'Coronary Artery Disease (CAD)', 'Hypothyroidism', 'Hyperlipidemia', 'Diabetes Mellitus (DM)', 'Height', 'Weight', 'Body Mass Index (BMI)', 'Total Body Water (TBW)', 'Extracellular Water (ECW)', 'Intracellular Water (ICW)', 'Extracellular Fluid/Total Body Water (ECF/TBW)', 'Total Body Fat Ratio (TBFR) (%)', 'Lean Mass (LM) (%)', 'Body Protein Content (Protein) (%)', 'Visceral Fat Rating (VFR)', 'Bone Mass (BM)', 'Muscle Mass (MM)', 'Obesity (%)', 'Total Fat Content (TFC)', 'Visceral Fat Area (VFA)', 'Visceral Muscle Area (VMA) (Kg)', 'Hepatic Fat Accumulation (HFA)', 'Glucose', 'Total Cholesterol (TC)', 'Low Density Lipoprotein (LDL)', 'High Density Lipoprotein (HDL)', 'Triglyceride', 'Aspartat Aminotransferaz (AST)', 'Alanin Aminotransferaz (ALT)', 'Alkaline Phosphatase (ALP)', 'Creatinine', 'Glomerular Filtration Rate (GFR)', 'C-Reactive Protein (CRP)', 'Hemoglobin (HGB)', 'Vitamin D'], dtype='object')

Show shape

```
df.shape
```

✓ 0.0s

(319, 39)

df.describe

count	method	NDFrame.describe of	Gallstone Status	Age	Gender	Comorbidity
0		0 50	0	0		
1		0 47	0	1		
2		0 61	0	0		
3		0 41	0	0		
4		0 42	0	0		
...		...	...	...		
314		1 49	0	0		
315		1 31	1	0		
316		1 58	0	0		
317		1 37	1	0		
318		1 60	1	0		
...		...	...	...		
0		Coronary Artery Disease (CAD)	Hypothyroidism	Hyperlipidemia		
1		0	0	0		
2		0	0	0		
3		0	0	0		
4		0	0	0		
...		...	...	...		
314		0	0	0		
315		0	0	0		
316		0	0	0		
317		0	0	0		
318		0	0	0		
...		...	...	...		
0		Diabetes Mellitus (DM)	Height	Weight	...	
1		0	185	92.8	...	
2		0	176	94.5	...	

- **Step 2: Missing Value Analysis**

A full check for missing values was done for all 38 feature variables and the target variable there were no missing values in the entire dataset. Missing values can create a biased model that will negatively impact the performance of the model, so identifying missing values is important to determine if filling missing value strategy is required. Which help the model to learn from true data distribution without being biased.

```
df.isnull().sum()
```

Gallstone Status	0
Age	0
Gender	0
Comorbidity	0
Coronary Artery Disease (CAD)	0
Hypothyroidism	0
Hyperlipidemia	0
Diabetes Mellitus (DM)	0
Height	0
Weight	0
Body Mass Index (BMI)	0
Total Body Water (TBW)	0
Extracellular Water (ECW)	0
Intracellular Water (ICW)	0
Extracellular Fluid/Total Body Water (ECF/TBW)	0
Total Body Fat Ratio (TBFR) (%)	0
Lean Mass (LM) (%)	0
Body Protein Content (Protein) (%)	0
Visceral Fat Rating (VFR)	0
Bone Mass (BM)	0
Muscle Mass (MM)	0
Obesity (%)	0
Total Fat Content (TFC)	0
Visceral Fat Area (VFA)	0
Visceral Muscle Area (VMA) (Kg)	0
...	
Glomerular Filtration Rate (GFR)	0
C-Reactive Protein (CRP)	0
Hemoglobin (HGB)	0
Vitamin D	0
dtype: int64	

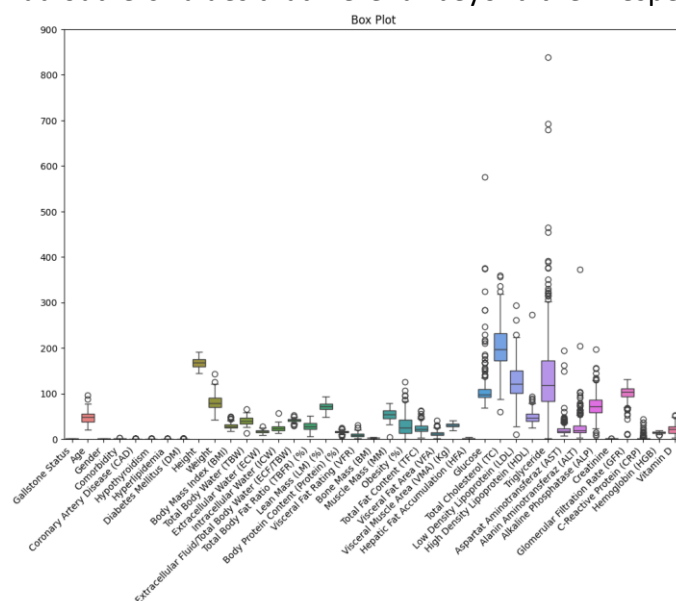
- **Step 3: Encoding Verification**

Numerical inputs are required for machine learning algorithms; verification is necessary in order to ensure that categorical variables have been encoded. Encoding matters because it directly affects how a model learns because many ML algorithms only accept numerical data (Udara, 2025). The dataset contains only Numerical data, so there is no need to use encoding methods.

df.dtypes	
Gallstone Status	int64
Age	int64
Gender	int64
Comorbidity	int64
Coronary Artery Disease (CAD)	int64
Hypothyroidism	int64
Hyperlipidemia	int64
Diabetes Mellitus (DM)	int64
Height	int64
Weight	float64
Body Mass Index (BMI)	float64
Total Body Water (TBW)	float64
Extracellular Water (ECW)	float64
Intracellular Water (ICW)	float64
Extracellular Fluid/Total Body Water (ECF/TBW)	float64
Total Body Fat Ratio (TBFR) (%)	float64
Lean Mass (LM) (%)	float64
Body Protein Content (Protein) (%)	float64
Visceral Fat Rating (VFR)	int64
Bone Mass (BM)	float64
Muscle Mass (MM)	float64
Obesity (%)	float64
Total Fat Content (TFC)	float64
Visceral Fat Area (VFA)	float64
Visceral Muscle Area (VMA) (Kg)	float64
Hepatic Fat Accumulation (HFA)	int64
Glucose	float64
Total Cholesterol (TC)	float64
Low Density Lipoprotein (LDL)	float64
High Density Lipoprotein (HDL)	float64
Triglyceride	float64
Aspartat Aminotransferaz (AST)	float64
Alanin Aminotransferaz (ALT)	float64
Alkaline Phosphatase (ALP)	float64
Creatinine	float64
Glomerular Filtration Rate (GFR)	float64
C-Reactive Protein (CRP)	float64
Hemoglobin (HGB)	float64
Vitamin D	float64
dtype: object	

- Step 4: Outliers Detecting and handling**

Outliers are extreme values of a dataset, which differ from the rest of the data points in a significant way. These Outliers can harm the machine learning model's performance in many ways like affecting the model's accuracy of prediction. To identify outliers, I used IQR a statistical method for determining how much variation exists between values in a set of data. Using IQR allows for outliers values to be determined using how much the middle 50 percent of the data varies. Any value outside the identified bounds is considered to be an extreme. A box plot was created to provide a visual representation of the data prior to any outlier handling. It was apparent upon reviewing the box plots that there were some features that had outliers values that were far beyond their respective whiskers.



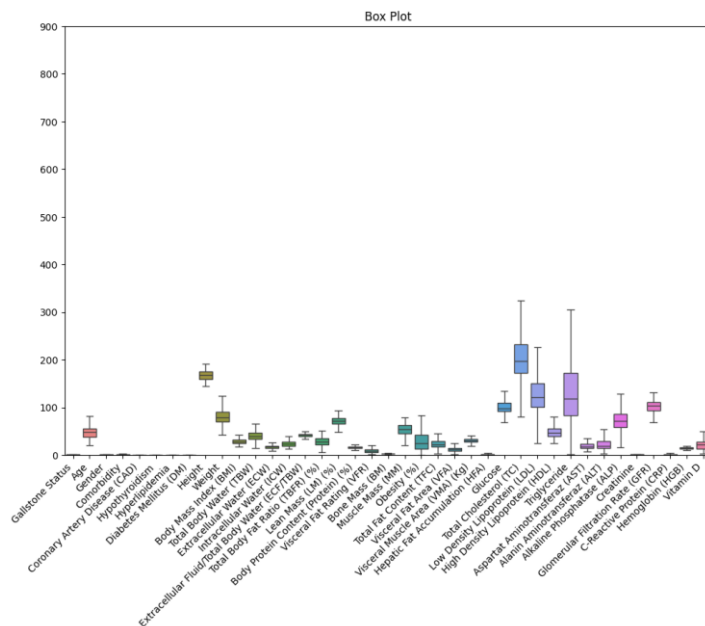
Clipping was the method used to handle the outliers. With this method, instead of removing the outliers from the data set, the outlier values are replaced by the boundary values that were previously identified using the IQR.

```
for i in df.columns:
    Q1= df[i].quantile(0.25)
    Q3= df[i].quantile(0.75)
    IQR= Q3-Q1
    LOWER= Q1 - 1.5*IQR
    UPPER= Q3 + 1.5*IQR

    df[i]= df[i].clip(lower=LOWER, upper=UPPER)
```

0.0s

A second box plot was created after clipping to verify that the outliers had been addressed. A programmatically verified loop also confirmed the absence of outliers, stating: "No Outliers".



Justification: Using clipping prevents the model from becoming skewed by extreme values, because SVM utilize distance-based method which is very sensitive to outliers. Moreover, Using clipping retains all 319 patient records. Retaining all of the patient records is especially important due to the small sample size.

- **Step 5: Train-Test-Split**

The dataset was split into training 80% and test 20% sets. To make sure that the Test Set has no influence in the Training Set, it is important to keep the Test Set completely separate from the training set so as to get a fair assessment of how well the model performs. Properly dividing the data into Train/Test Sets prevents Data Leakage and provide a Reliable Estimates of models Performance.

Split data to features and target

```
X=df.drop(['Gallstone Status'], axis=1)
y=df['Gallstone Status']
```

✓ 0.0s

Split the data into training and testing sets

```
X_train, X_test, y_train, y_test= train_test_split(X, y, test_size=0.2, random_state=42)
```

✓ 0.0s

- **Step 6: Feature Scaling**

It is very important to first scale all data before apply machine learning models, because many models will not function at their best. Models such as Support Vector Machine rely heavily upon the scale of the input features, so it will perform poorly. This is because models that calculate distance between data points (like SVM) will tend to weigh the input features differently based upon their scale. So, applying Standard Scaling will ensure that all the input features contribute equally to the model.

On the other hand, Decision tree, Random forest, XGBoost, and Gradient Boosting do not need to be scaled. These types of models are based upon splitting the data into separate regions based upon the value of each feature.

Feature Scaling

```
Scaler=StandardScaler()
X_train_Scaled= Scaler.fit_transform(X_train)
X_test_Scaled= Scaler.transform(X_test)
```

✓ 0.0s

2 Model Implementation (Q2)

Table 1: Description of the hyperparameters and values considered.

Algorithm	Hyperparameter	Description	Values considered
SVM	C	Regularization parameter which controls the trad-off between low training error and low testing error	0.1, 1, 10
SVM	Kernal	Choose the Kernel type that will be used in the algorithm	Linear, rbf
SVM	Gamma	Kernel coefficient for rbf kernel which defines how far the influence of a single training example reaches	Scale, auto
Random Forest	n_estimators	Number of trees in the Forest	50, 100, 150
Random Forest	max_depth	Maximum depth number for each tree	5, 10, 20
Random Forest	criterion	The function that measure the quality of a split	gini , entropy
XGBoost	learning_rate	Reduce the step size that used to prevent overfitting	0.01, 0.1, 0.2
XGBoost	max_depth	Maximum depth number for each tree	3, 6, 9
XGBoost	n_estimators	Number of rounds (boosting)	50, 100, 200
Gradient Boosting	learning_rate	Reduce the contribution of each tree	0.01, 0.1, 0.2
Gradient Boosting	max_depth	Maximum depth of the individual regression estimators	3, 5, 7
Gradient Boosting	n_estimators	Number of boosting stages to be implemented	50, 100, 200
Decision Tree	max_depth	Maximum depth of the tree	5, 10, 20
Decision Tree	criterion	The function that measure the quality of a split	gini , entropy

Table 2: Combination of the hyperparameter values and corresponding performance achieved.

Algorithm	Combination of hyperparameter values	Training performance	Validation performance
SVM	{'C': 0.1, 'gamma': 'scale', 'kernel': 'linear'}	0.8470	0.7960
SVM	{'C': 0.1, 'gamma': 'auto', 'kernel': 'linear'}	0.8470	0.7960
SVM	{'C': 10, 'gamma': 'auto', 'kernel': 'linear'}	0.8686	0.7843
SVM	{'C': 1, 'gamma': 'auto', 'kernel': 'rbf'}	0.8901	0.7921
SVM	{'C': 1, 'gamma': 'scale', 'kernel': 'linear'}	0.8607	0.7803
	<pre> SVM Top 10 Combanations: params mean_train_score \ 0 {'C': 0.1, 'gamma': 'scale', 'kernel': 'linear'} 0.847059 2 {'C': 0.1, 'gamma': 'auto', 'kernel': 'linear'} 0.847059 7 {'C': 1, 'gamma': 'auto', 'kernel': 'rbf'} 0.890196 5 {'C': 1, 'gamma': 'scale', 'kernel': 'rbf'} 0.900980 10 {'C': 10, 'gamma': 'auto', 'kernel': 'linear'} 0.868627 4 {'C': 1, 'gamma': 'scale', 'kernel': 'linear'} 0.860784 9 {'C': 10, 'gamma': 'scale', 'kernel': 'rbf'} 0.997059 11 {'C': 10, 'gamma': 'auto', 'kernel': 'rbf'} 0.994118 3 {'C': 0.1, 'gamma': 'auto', 'kernel': 'rbf'} 0.698039 1 {'C': 0.1, 'gamma': 'scale', 'kernel': 'rbf'} 0.699020 mean_test_score 0 0.796078 2 0.796078 7 0.792157 5 0.788235 10 0.784314 4 0.780392 9 0.764706 11 0.764706 3 0.639216 1 0.631373 </pre>		
Random Forest	{'criterion': 'entropy', 'max_depth': 5, 'n_estimators': 100}	0.9656	0.7921
Random Forest	{'criterion': 'entropy', 'max_depth': 10, 'n_estimators': 150}	1.0	0.7843
Random Forest	{'criterion': 'gini', 'max_depth': 10, 'n_estimators': 150}	1.0	0.7803
Random Forest	{'criterion': 'gini', 'max_depth': 5, 'n_estimators': 50}	0.9754	0.7803
Random Forest	{'criterion': 'entropy', 'max_depth': 5, 'n_estimators': 50}	0.9715	0.7607

	<pre> Random Forest Top 10 Combanations: params \ 10 {'criterion': 'entropy', 'max_depth': 5, 'n_estimators': 100} 14 {'criterion': 'entropy', 'max_depth': 10, 'n_estimators': 150} 5 {'criterion': 'gini', 'max_depth': 10, 'n_estimators': 150} 0 {'criterion': 'gini', 'max_depth': 5, 'n_estimators': 50} 12 {'criterion': 'entropy', 'max_depth': 10, 'n_estimators': 50} .. 9 {'criterion': 'entropy', 'max_depth': 5, 'n_estimators': 50} 4 {'criterion': 'gini', 'max_depth': 10, 'n_estimators': 100} 7 {'criterion': 'gini', 'max_depth': 20, 'n_estimators': 100} 16 {'criterion': 'entropy', 'max_depth': 20, 'n_estimators': 100} 6 {'criterion': 'gini', 'max_depth': 20, 'n_estimators': 50} mean_train_score mean_test_score 10 0.965686 0.792157 14 1.000000 0.784314 5 1.000000 0.780392 0 0.975490 0.780392 12 1.000000 0.776471 9 0.971569 0.760784 4 1.000000 0.760784 7 1.000000 0.749020 16 1.000000 0.745098 6 1.000000 0.737255 </pre>		
XGBoost	<code>{'learning_rate': 0.2, 'max_depth': 9, 'n_estimators': 200}</code>	1.0	0.7764
XGBoost	<code>{'learning_rate': 0.01, 'max_depth': 6, 'n_estimators': 200}</code>	0.9931	0.7176
XGBoost	<code>{'learning_rate': 0.01, 'max_depth': 6, 'n_estimators': 100}</code>	0.9598	0.7098
XGBoost	<code>{'learning_rate': 0.01, 'max_depth': 6, 'n_estimators': 50}</code>	0.9343	0.7098
XGBoost	<code>{'learning_rate': 0.01, 'max_depth': 9, 'n_estimators': 50}</code>	0.9431	0.7019
	<pre> XGBoost Top 10 Combanations: params \ 26 {'learning_rate': 0.2, 'max_depth': 9, 'n_estimators': 200} 25 {'learning_rate': 0.2, 'max_depth': 9, 'n_estimators': 100} 20 {'learning_rate': 0.2, 'max_depth': 3, 'n_estimators': 200} 17 {'learning_rate': 0.1, 'max_depth': 9, 'n_estimators': 200} 21 {'learning_rate': 0.2, 'max_depth': 6, 'n_estimators': 50} .. 5 {'learning_rate': 0.01, 'max_depth': 6, 'n_estimators': 200} 4 {'learning_rate': 0.01, 'max_depth': 6, 'n_estimators': 100} 3 {'learning_rate': 0.01, 'max_depth': 6, 'n_estimators': 50} 6 {'learning_rate': 0.01, 'max_depth': 9, 'n_estimators': 50} 7 {'learning_rate': 0.01, 'max_depth': 9, 'n_estimators': 100} mean_train_score mean_test_score 26 1.000000 0.776471 25 1.000000 0.768627 20 1.000000 0.768627 17 1.000000 0.764706 21 1.000000 0.764706 5 0.993137 0.717647 4 0.959804 0.709804 3 0.934314 0.709804 6 0.943137 0.701961 7 0.974510 0.698039 </pre>		
Gradient Boosting	<code>{'learning_rate': 0.2, 'max_depth': 5, 'n_estimators': 200}</code>	1.0	0.7647
Gradient Boosting	<code>{'learning_rate': 0.1, 'max_depth': 3, 'n_estimators': 50}</code>	0.9980	0.7529

Gradient Boosting	{'learning_rate': 0.1, 'max_depth': 5, 'n_estimators': 100}	1.0	0.7490
Gradient Boosting	{'learning_rate': 0.1, 'max_depth': 5, 'n_estimators': 200}	1.0	0.7450
Gradient Boosting	{'learning_rate': 0.2, 'max_depth': 3, 'n_estimators': 200}	1.0	0.7450
	<pre> Gradient Boosting Top 10 Combanations: params \ 23 {'learning_rate': 0.2, 'max_depth': 5, 'n_estimators': 200} 9 {'learning_rate': 0.1, 'max_depth': 3, 'n_estimators': 50} 13 {'learning_rate': 0.1, 'max_depth': 5, 'n_estimators': 100} 14 {'learning_rate': 0.1, 'max_depth': 5, 'n_estimators': 200} 20 {'learning_rate': 0.2, 'max_depth': 3, 'n_estimators': 200} .. 24 {'learning_rate': 0.2, 'max_depth': 7, 'n_estimators': 50} 15 {'learning_rate': 0.1, 'max_depth': 7, 'n_estimators': 50} 17 {'learning_rate': 0.1, 'max_depth': 7, 'n_estimators': 200} 8 {'learning_rate': 0.01, 'max_depth': 7, 'n_estimators': 200} 7 {'learning_rate': 0.01, 'max_depth': 7, 'n_estimators': 100} mean_train_score mean_test_score 23 1.000000 0.764706 9 0.998039 0.752941 13 1.000000 0.749020 14 1.000000 0.745098 20 1.000000 0.745098 .. 24 1.000000 0.682353 15 1.000000 0.678431 17 1.000000 0.666667 8 1.000000 0.662745 7 1.000000 0.658824 </pre>		
Decision Tree	{'criterion': 'entropy', 'max_depth': 5}	0.9088	0.7137
Decision Tree	{'criterion': 'entropy', 'max_depth': 10}	0.9960	0.6941
Decision Tree	{'criterion': 'entropy', 'max_depth': 20}	1.0	0.6784
Decision Tree	{'criterion': 'gini', 'max_depth': 5}	0.9254	0.6588
Decision Tree	{'criterion': 'gini', 'max_depth': 20}	1.0	0.6470
Decision Tree	{'criterion': 'gini', 'max_depth': 10}	0.9970	0.6235

3 Model Testing and Evaluation (Q3)

A variety of performance metrics were used to assess model performance in the context of predicting gallstone disease, each addressing a different aspect of model performance.

Selected Evaluation Metrics:

- 1- **Accuracy:** Accuracy measures the total number of accurate predictions (both true positives and true negatives) divided by the total number of predictions made. Accuracy is suitable for this problem since there is near parity in the distribution of the data (50.5% of the patients have gallstones and 49.5% do not); therefore, accuracy will provide a reasonable measure of total model performance and will be free from bias resulting from class imbalance.

$$\text{Accuracy} = (TP + TN) / (TP + TN + FP + FN)$$

- 2- **Precision:** Precision measures the ratio of accurately predicted positive cases (patients with gallstones) to the total number of cases that the model predicts are positive. Precision is suitable for medical problems because it is a significant factor in medical diagnosis since it represents the reliability of the model in identifying patients as having gallstones. Higher precision levels indicate fewer false positives, further testing, and medical expense for patients who do not truly have the disease.

$$\text{Precision} = TP / (TP + FP)$$

- 3- **Recall:** Recall measures the percentage of actual positive cases (patients with gallstones) that were correctly identified by the model. Recall is suitable for medical problems because it is very important in medical applications since it represents the model's ability to find all patients who have gallstones. High levels of recall ensure that fewer patients with the disease are missing (false negatives), which is critical since undiagnosed gallstones may result in serious complications.

$$\text{Recall} = TP / (TP + FN)$$

- 4- **F1 Score:** F1 score is the compatible mean of precision and recall and represents a balanced measure of both. F1 is useful for balancing the trade-offs between precision and recall. In healthcare, we would like to both minimize false positives (to prevent unnecessary treatments) and minimize false negatives (to prevent missing actual cases). The F1 score provides a single metric that captures this balance.

$$F1 = 2 * (\text{Precision} * \text{Recall}) / (\text{Precision} + \text{Recall})$$

- 5- **Confusion Matrix:** A confusion matrix represents a complete breakdown of the model's predictions, which includes:

- a- True Negative: Cases that were correctly predicted as not having gallstones.
- b- True Positive: Cases correctly predicted as having gallstones.
- c- False Negative: Cases incorrectly predicted as not having gallstones.
- d- False Positive: Cases incorrectly predicted as having gallstones.

Confusion matrix is suitable for medical problems because it provides insight into the model's errors and serves as the basis for calculation of the other metrics.

$$\text{Confusion Matrix} = \begin{bmatrix} TN & FP \\ FP & TP \end{bmatrix}$$

Why These Metrics were selected:

- Address Medical Context: Both false positives and false negatives in gallstone prediction have clinically relevant implications. False positives will require additional imaging studies and cause unnecessary worry for patients. False negatives will lead to delayed treatment and potentially serious complications for those patients with gallstones who remain untreated.
- Provide Comprehensive Evaluation: Multiple metrics allow for a comprehensive view of model performance and enable informed decisions regarding which model should be deployed clinically.
- Address Balanced Data Set: Since the data set is roughly balanced, accuracy provides meaningful information that is not misleading due to the class imbalance.

Table 3: Summary of the test results.

Algorithm	Best combination of hyperparameters	Test accuracy	Test precision	Test recall	Test F1	Confusion Matrix [TN, FP] [FP, TP]
SVM	{'C': 0.1, 'gamma': 'scale', 'kernel': 'linear'}	0.875	0.8965	0.8387	0.86666	[[30, 3], [5, 26]]
Random Forest	{'criterion': 'entropy', 'max_depth': 10, 'n_estimators': 50}	0.8125	0.8275	0.7742	0.8	[[28, 5], [7, 24]]
XGBoost	{'learning_rate': 0.2, 'max_depth': 9, 'n_estimators': 200}	0.8437	0.8387	0.8387	0.8387	[[28, 5], [5, 26]]
Gradient Boosting	{'learning_rate': 0.1, 'max_depth': 5, 'n_estimators': 200}	0.8281	0.8571	0.7742	0.8135	[[29, 4], [6, 25]]
Decision Tree	{'criterion': 'entropy', 'max_depth': 5}	0.7656	0.7857	0.7096	0.7457	[[27, 6], [9, 22]]

4 Bias and Variance (Q4)

- Bias (Underfitting): It means that the model is too simple to capture the patterns in the data, so the result is a model with poor performance on both training and testing sets.
- Variance (Overfitting): It means that that model is too complex, so the model learn noise, memorize the data not learning, and don't generalize to new data. So, the result is a model with excellent training performance but very poor testing performance.
- Good fit: It means that the model achieves a balance between bias and variance, performing well on training and testing sets with similar performance metrics.

To identify if the model is overfitting, underfitting, or good fit, I analyze model performance:

1- SVM:

SVM is a Good fit model. The best combination was $C=0.1$, $\gamma=\text{scale}$, $\text{kernel}=\text{linear}$.

- *The Cross validation performance analysis:*

The best combination gives training score= 0.847 which is bigger than validation score= 0.796, So, the gap between them is 0.051, this small gap indicate that the model learns consistent patterns without memorizing training data.

- *Training and Testing performance analysis:*

The best combination gives training accuracy= 85.1% which is smaller than testing accuracy= 87.5%, So, the test accuracy exceeds training accuracy by 2.4% which is a strong indicator that the model generalized excellently.

- *Precision, Recall, F1-score analysis:*

The best combination gives Precision= 89.66%, Recall= 83.87%, and $F1=86.67\%$, these all metrics are well balanced and high scores. Precision and Recall both are above 80% which indicates that the model performs reliably in both categories without bias.

- *Confusion Matrix analysis:*

The best combination gives True Negative=30, False Positive=3, False Negative=5, and True Positive=26, so, the total correct predictions=56/64. The confusion matrix shows only 8 errors (FN+FP) that distributed quite evenly between the two groups. This confirms consistent and reliable predictions.

From all aspects SVM shows that model complexity is just enough to capture patterns (87.5% test accuracy) which gives low bias, Stable performance across CV which gives low variance, and it achieves optimal balance without overfitting or underfitting. The SVM model is generally the best and is recommended for use in clinical applications.

2- Random Forest:

Random Forest is an Overfitting model. The best combination was criterion= entropy, max_depth= 5, n_estimators= 100.

- *The Cross validation performance analysis:*

The best combination gives training score= 1.00 which is extremely bigger than validation score= 0.792, So, the gap between them is 0.207 (20%), this large gap indicate that the model does not learn generalizable patterns, it memorizes training data.

- *Training and Testing performance analysis:*

The best combination gives training accuracy= 100% which is much bigger than testing accuracy= 81.25%, So, the training accuracy exceeds testing accuracy by 18.75% which is a strong indicator of overfitting. The model only learned training patterns which don't transferred to the testing patients' data (new data).

- *Precision, Recall, F1-score analysis:*

The best combination gives Precision= 82.76%, Recall= 77.42%, and F1=80.00%, will these test metrics are reasonable but they are lower than the training metrics (all 100%), which confirms the generalization gap and overfitting.

- *Confusion Matrix analysis:*

The best combination gives True Negative=28, False Positive=5, False Negative=7, and True Positive=24, so, the total correct predictions=52/64. The confusion matrix shows 12 errors (FN+FP). The 7 FN are particularly concerning for medical diagnosis because they represent undetected gallstones.

From all aspects Random Forest shows that model complexity is too high and the model memorizes training data 100% accuracy which gives low bias, unstable performance with large CV gap 20.78% and test gap 18.75% that gives high variance, so, it fail in generalization and causing overfitting, RF is not suitable for clinical problems.

3- XGBoost:

XGBoost is an Overfitting model. The best combination was learning_rate= 0.2, max_depth= 9, n_estimators= 200.

- *The Cross validation performance analysis:*

The best combination gives training score= 1.00 which is extremely bigger than validation score= 0.7765, So, the gap between them is 0.223 (22.3%), this is the largest gap between all the models, this gap indicate that the model overfits more than Random Forest and SVM during cross-validation.

- *Training and Testing performance analysis:*

The best combination gives training accuracy= 100% which is much bigger than testing accuracy= 84.38%, So, the training accuracy exceeds testing accuracy by 15.62% which is a strong indicator of overfitting. Even if the

testing performance is better than Random Forest, the model still memorizes the training patterns rather than learning generalizable patterns.

- *Precision, Recall, F1-score analysis:*

The best combination gives Precision= 83.87%, Recall= 83.87%, and F1=83.87%, Precision and recall are equivalent this indicating a very good balance of model performance. However, the large gap between training performance (100%), indicates an extreme amount of overfitting and poor generalization.

- *Confusion Matrix analysis:*

The best combination gives True Negative=28, False Positive=5, False Negative=5, and True Positive=26, so, the total correct predictions=54/64. The confusion matrix shows 10 errors (FN+FP) which means that the model does equally poorly to both classes when presented with new data as evidenced by the balanced distribution of errors. This represents that the model is better than Random Forest but poorer than SVM.

From all aspects XGBoost demonstrates that this model is extremely complex and achieves perfect training (100%), therefore XGBoost has low bias, high variance, and extreme overfitting with the largest cv gap 22.35% and significant test gap 15.62%. As a result, XGBoost outperforms random forest on test accuracy so it needs much stronger regularization and shouldn't be used clinically in this configuration.

4- Gradient Boosting:

Gradient Boosting is an Overfitting model. The best combination was learning_rate= 0.1, max_depth= 5, n_estimators= 200.

- *The Cross validation performance analysis:*

The best combination gives training score= 1.00 which is much bigger than validation score= 0.7529, So, the gap between them is 0.247 (24.7%). This large gap indicates that the model has the most severe overfitting during training.

- *Training and Testing performance analysis:*

The best combination gives training accuracy= 100% which is much bigger than testing accuracy= 82.81%, So, the training accuracy exceeds testing accuracy by 17.1% which is the second-largest test gap. This confirms the overfitting behaviour across the different stages of evaluation.

- *Precision, Recall, F1-score analysis:*

The best combination gives Precision= 85.71%, Recall= 77.42%, and F1=81.36%, will these test metrics be reasonable, but they are lower than the training metrics (all 100%), which confirms the generalization gap and overfitting. Although the model has a relatively good precision score, it also shows a poor ability to capture actual gallstone positives (recall), resulting in an unacceptable number of cases being missed by the model which could have serious implications for the accurate use of this model

in clinical diagnosis where the importance of capturing True positives can't be overstated.

- *Confusion Matrix analysis:*

The best combination gives True Negative=29, False Positive=4, False Negative=7, and True Positive=24, so, the total correct predictions=53/64. The confusion matrix shows 11 errors (FN+FP). The 7 FN are particularly concerning for medical diagnosis because they represent undetected gallstones and incorrectly classified as healthy.

From all aspects Gradient Boosting shows that model complexity is too high and a perfect model training performance 100%, low bias, the largest cv gap 24.71%, second-largest test gap 17.19%, and extreme overfitting potential which would be a very poor candidate for clinical use because of the significant risk

5- Decision Tree:

Random Forest is an Overfitting model with the worst overall performance. The best combination was criterion= entropy, max_depth= 5.

- *The Cross validation performance analysis:*

The best combination gives training score= 0.9088 which is bigger than validation score= 0.721, So, the gap between them is 0.1872 (20%), this large difference shows that the model is still overfitting, indicating that decision trees are naturally vulnerable to overfitting even with limited depth.

- *Training and Testing performance analysis:*

The best combination gives training accuracy= 88.24% which is bigger than testing accuracy= 76.56%, So, the training accuracy exceeds testing accuracy by 11.68% which indicates a moderate overfitting. Moreover, the model shows the lowest test accuracy among all the models.

- *Precision, Recall, F1-score analysis:*

The best combination gives Precision= 78.57%, Recall= 70.96%, and F1=74.58%, these test metrics are lower than all other models. The model has poor sensitivity if the recall is below 80%. This means that the model will miss about thirty percent of actual gallstone cases which would be unsuitable in a clinical setting.

- *Confusion Matrix analysis:*

The best combination gives True Negative=27, False Positive=6, False Negative=9, and True Positive=22, so, the total correct predictions=49/64. The confusion matrix shows 15 errors (FN+FP). The 9 FN indicates that 29% of actual positive cases didn't detect, that shows the lower reliability for clinical use.

From all aspects Decision tree shows that model does not have sufficient complexity to capture patterns because it has the minimum test accuracy at 76.56% that gives a moderate bias and remains overfitted despite its shallow depth with a CV gap of 18.72%. and a test gap of 11.68%, resulting in high variance. It exhibits both underfitting

(lacks power) and overfitting (fails to generalize), resulting in the worst overall performance with the highest error rate 23.44%. The decision tree is not recommended for clinical use.

5 Model effectiveness analysis (Q5)

- *Analyse the results:*

1. **SVM:**

The SVM model gives 87.5% accuracy. This indicates that 87.5% of all patients in the test set were correctly classified as suffering from gallstones or not. In any medical context, the high accuracy suggests that the SVM model is highly accurate in predicting gallstones disease which reduces the risk of having misdiagnosed gallstones and unnecessary costly medical tests.

2. **XGBoost:**

The XGBoost model gives 84.37% accuracy. This demonstrates good predictive ability as most patients were classified correctly. Although there was little difference to the results from the SVM; this suggests that despite XGBoost having good strengths (predictive) it may have a tendency to over-fit (complexity), resulting in incorrect predictions on new patient data.

3. **Gradient Boosting:**

The Gradient Boosting model gives 82.81% accuracy. This accuracy reflects good performance, but it is less than XGBoost and SVM. In a clinical setting, the model's lack of precision will mean it remains a viable tool; however, this will result in an increased number of incorrect classifications compared to both the XGBoost and SVM models, which could delay diagnoses for certain patients.

4. **Random Forest:**

The Random Forest model gives 81.25% accuracy. This accuracy reflects that the random forest model is good but less effective than SVM and Boosting models. For gallstone diagnosis, this means more false positives or false negatives, which could lead to unnecessary imaging or missed diagnoses.

5. **Decision Tree:**

The Decision Tree model gives 76.56% accuracy. This accuracy is the lowest among all models; this means that 23.44% of predictions were wrong (nearly one in four patients misclassified). This resulted in nearly 1 in every 4 patients being classified incorrectly (23.44% of all classifications made by the model), which represents a high degree of misclassification. This amount of error would be unacceptable in a clinical context as it results in serious misdiagnosis.

- *Compare the algorithms based on these accuracy test results:*

SVM showed the best results (accuracy = 87.5%) to indicate that it has a good balance of complexity and generalization ability and therefore was able to generalize from the training data well and capture the patterns in the data. The second result came from XGBoost with an accuracy of 84.37%. The results from XGBoost suggest high predictive capabilities for this model; however, there were also signs of overfitting since the model's complexity is higher than the other models. Both

Gradient Boosting and Random Forest gave similar results (accuracy = 82.81% and 81.25%), respectively, to show that both are very robust models, but have slightly less generalization as compared to both the results from SVM and XGBoost. The worst performing model was the Decision Trees with the lowest accuracy (76.56%). These results were expected, since decision trees have no ensemble strength and they are typically overfit the data.

- *Alignment between the expected capabilities and the results*

1- SVM:

In many ways, the SVM works well with the binary classification problem. A lot of times, this is when the data is balanced, so that all of the relevant data points can be taken into account during the learning process and when scaling of features has occurred. These two things show us that the SVM's best performance shows that we will likely be able to generalize well from the model with little to no risk of overfitting the model, which makes it a good choice for use in medical prediction applications.

2- XGBoost:

XGBoost is known for being able to deal with complicated relationships in the data as part of its boosting. This is why we would expect it to have a strong degree of predictive power because of the way it learns based on its algorithm. Although the difference between how the model performs using our test set versus our training set is relatively small, the difference does indicate some amount of overfitting, which is typical for most of the boosting-based models of machine learning, but this type of overfitting could potentially be avoided if the hyperparameters used in the model were to be closely controlled.

3- Gradient Boosting:

Gradient Boosting is generally a very effective method; however, it tends to be more vulnerable to overfitting than XGBoost. Therefore, its accuracy represents good overall performance, but is slightly lower than XGBoost, indicating the theoretical differences between them as forms of boosting, and the simplicity of XGBoost.

4- Random Forest:

Random Forest is expected to produce accurate and robust results, but as an ensemble method, it does not have the same degree of interactivity in terms of feature complexity as boosting methods. As such, its slightly lower accuracy compared to XGBoost and Gradient Boosting is entirely consistent with what would be expected given the strengths of each type of method.

5- Decision Tree:

Decision Trees are generally subject to overfitting and lack the ensemble strength of Random Forest and the boosting methods previously mentioned. This was clearly demonstrated by the decision tree achieving the lowest accuracy, and therefore confirming the limitations associated with using a single decision tree, as opposed to an ensemble of multiple decision trees.

6 Model effectiveness evaluation (Q6)

6.1 Effectiveness evaluation

- SVM:

Metric	Value	Evaluation
Accuracy	0.875	The model has accurately classified 87.5% of the patients. This high accuracy of the model indicated that the model is capable of strong generalization of unseen data, as a result, SVM is a very viable model of classifying Gallstone Disease.
Precision	0.8965	Precision of 89.65 means that the model can be trusted to predict the patient that has gallstone because it is accurately predict 9 out of 10 patients, this reduces unnecessary medical testing and anxiety for patients.
Recall	0.8387	Recall of 83.87% means that the model can identify almost all gallstone patients, which is critical in medical diagnostics so as not miss actual gallstone patients. Because if the model misses a lot of cases this will cause delays in detecting the disease. SVM shows the highest recall value.
F1-Score	0.8667	A F1-score of 86.67% means that the model balances between Precision and Recall well thereby providing reliable predictions and minimizing False Positive (Patients who were incorrectly classified as having gallstones when they did not), and False Negative (Patients who were incorrectly classified as not having gallstones when they did).
Confusion Matrix	[30, 3] [5, 26]	The confusion matrix shows that there were 30 patients who were correctly classified as not having stones, 3 patients who were incorrectly classified as having stones when they did not, 5 patients who were incorrectly classified as not having stones when they did, and 26 patients who were correctly classified as not having stones. In total, only 8 errors occurred therefore the SVM model is extremely reliable and has low misclassification rate.

In terms of performance overall, the SVM model clearly shows strong results in all metrics (accuracy, precision, recall, and F1-score). The SVM model can achieve a good balance of correct identification of gallstone patients with a minimal number of false positives. The confusion matrix shows that the SVM model also has a very low error rate, which is important when using the model for clinical purposes. Therefore, SVM would be the most reliable option for this specific medical prediction task.

- **Random Forest:**

Metric	Value	Evaluation
Accuracy	0.8125	The model classifier has an accuracy of 81.25%. The model performed fairly well, however, it was less accurate than the SVM model.
Precision	0.8275	Precision of 82.75% means that the model identifies most of the patients with gallstones as having gallstones, and therefore the model will rarely suggest an unnecessary procedure due to a false positive.
Recall	0.7742	Recall of 77.42% means the model fails to identify some patients with gallstones, and this would be a problem in healthcare because it could result in some gallstones going undiagnosed.
F1-Score	0.8000	F1 score of 80.00% means that the Random Forest model performs equally well on both precision and recall metrics, so, the model does not perform as well as the SVM model at finding a balance between precision and recall.
Confusion Matrix	[28, 5] [7, 24]	The confusion matrix for this model is 28 patients who were correctly classified as not having stones, 5 patients who were incorrectly classified as having stones when they did not, 7 patients who were incorrectly classified as not having stones when they did, and 24 patients who were correctly classified as having stones. In total, 12 errors occurred. So, the Random Forest model is moderately good at detecting true positives; however, the number of false negatives may be a problem when dealing with medical diagnoses.

Random forest provides moderate results but the inability to provide a high recall score this means that the model misses gallstone patients. Although the model's precision is acceptable, the low level of recall is a major concern in clinical use because every positive case needs to be identified. Additionally, the Random Forest model demonstrates evidence of overfitting, thus it has reduced ability to generalize to new data.

- **XGBoost:**

Metric	Value	Evaluation
Accuracy	0.8437	The model has accurately classified 84.37% of the patients. This indicate hight reliable, however, the accuracy of the XGBoost is slightly lower than SVM but slightly higher than Random Forest.
Precision	0.8387	The precision of 83.87% XGBoost showed that the model produced highly reliable predictions regarding the presence of gallstones; However, similar to Random Forest, it is likely that it led to many unnecessary diagnostic tests for patients who were incorrectly predicted to have gallstones, resulting in unnecessary financial costs and psychological distress. Therefore, SVM outperforms XGBoost because it is more accurate in detecting and correctly classifying non gallstone patients.
Recall	0.8387	Also, the XGBoost gives recall of 83.87%, showed that the model produced highly reliable predictions regarding the presence of gallstones; however, similar to Random Forest, it is possible that it did not classify those with gallstones as affected, leading to a delay in the detection and treatment of the disease. Therefore, XGBoost outperformed Random Forest, but SVM still outperformed XGBoost because it was more accurate in detecting and correctly classifying no gallstone.
F1-Score	0.8387	An F1-score of 83.87% for XGBoost demonstrates a balanced ability to predict Gallstones without biasing either Precision or Recall.
Confusion Matrix	[28, 5] [5, 26]	The Confusion Matrix had 28 patients who were correctly classified as not having stones, 5 patients who were incorrectly classified as having stones when they did not, 5 patients who were incorrectly classified as not having stones when they did and 26 patients who were correctly classified as having stones. The distribution of errors was relatively even, showing that the model performed about equally well across both classes, but there was still potential for additional improvement.

In regard to both Precision and Recall, XGBoost is a very effective model for identifying Gallstones. Additionally, due to the potential for overfitting, XGBoost would be expected to have less accuracy on unseen data, although it would still be a reliable model for use in

prediction-based applications. XGBoost is better than Random Forest, but SVM will remain the best choice for clinical application.

- **Gradient Boosting:**

Metric	Value	Evaluation
Accuracy	0.8281	Gradient Boosting was able to achieve an average of 82.81% accuracy, a very good performance; although this is slightly lower than the other two models (XGBoost & SVM), they were all fairly accurate at predicting whether a patient had gallstones or not.
Precision	0.8571	Precision was also very good at 85.71% which means that the model did a good job of identifying those gallstones it predicted would be present as positive.
Recall	0.7742	Recall of 77.42% means that the model missed some gallstone cases and therefore produced false negatives, a major drawback in medical applications where finding all possible cases is crucial
F1-Score	0.8135	An F1-score of 81.35% suggests a good balance, however, is not as effective as SVM for overall effectiveness.
Confusion Matrix	[29, 4] [6, 25]	The confusion matrix shows that there are 29 patients who were correctly classified as not having stones, 4 patients who were incorrectly classified as having stones when they did not, 6 patients who were incorrectly classified as not having stones when they did, and 25 patients who were correctly classified as having stones. The data indicate that while the model has a great deal of strength, it still misses a small number of gallstone cases, a serious consideration for medical applications.

Gradient Boosting performed with a lot of strength in terms of precision, but was weak in recall, missing some positive cases. The F1-score suggests a good balance between the two, but the fact that the model tends to miss some gallstone patients makes it somewhat suboptimal for clinical application. There is still the risk of overfitting; however, it performed better than both Random Forest and Decision Trees.

- **Decision Tree:**

Metric	Value	Evaluation
Accuracy	0.7656	Decision Trees had the least accurate results of 76.56%, showing that it may be less reliable than the other models tested to accurately classify patient's conditions with respect to gallstones.
Precision	0.7857	The Precision score of 78.57% indicates that while the model is somewhat able to correct identify gallstones, there are still many false positive classifications.
Recall	0.7096	The Recall score of 70.96% was the least of all models tested, showing that the model missed many of the actual gallstone cases, which can be problematic when making medical diagnoses.
F1-Score	0.7457	The F1 Score of 74.57% indicates that the model is biased toward Precision (correctly identifying gallstones), but at the expense of Recall (missing actual gallstone cases). This bias could potentially lead to some actual gallstone cases going undiagnosed.
Confusion Matrix	[27, 6] [9, 22]	The confusion matrix shows that there are 27 patients who were correctly classified as not having stones, 6 patients who were incorrectly classified as having stones when they did not, 9 patients who were incorrectly classified as not having stones when they did, and 23 patients who were correctly classified as having stones. In total, this model gave the worst number of errors which is 15. The Decision Trees model is also unsuitable for medical use due to its inability to correctly diagnose many gallstone cases.

Overall, the Decision Trees Model performed the worst on all metrics. Its low Accuracy and low Recall rates show that it failed to correctly identify a significant number of gallstone patients. The Confusion Matrix showed the most False Negatives of any of the models tested, and therefore the Decision Trees model is not recommended for any clinical or medical applications.

6.2 Explainable ML techniques

To improve transparency and trust of ML models, I applied two explainable AI techniques **Shap and Lime**. Shap is a global model interpretation and Lime is an individual prediction explanation.

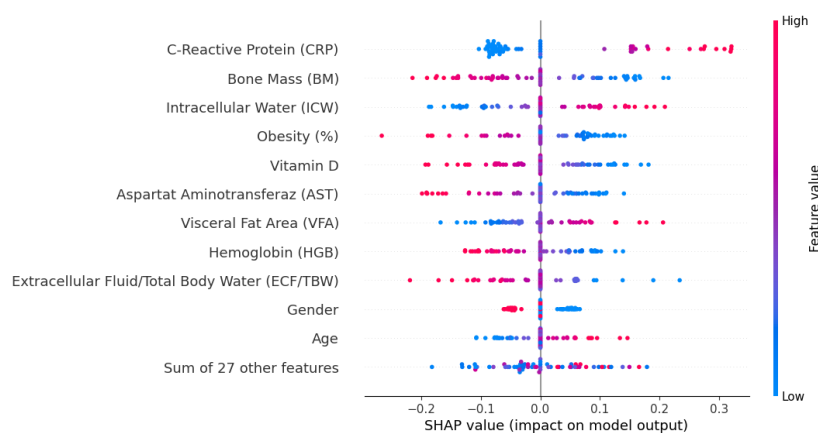
What Shap plot shows:

- **The negative values: Decrease gallstone risk “No gallstone increase”**
- **The Positive values: Increase gallstone risk “gallstone increase”**
- **Colour Coding: Red [high feature values “High probability”], Blue [low feature values]**
- **Dot: each dot is one patient**
- **The x-axis: represent the Shap values (the impact on the prediction).**
- **The y-axis: represents the features that are ranked by the importance.**

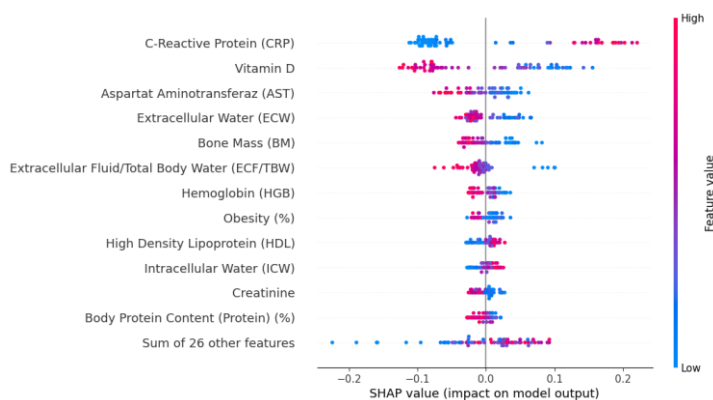
The Shap explains models at a global level, by showing how each feature impacts predictions in the models. The Shap Summary Plot Ranks the Most Important Features based on their overall impact on the Output of the Model. Features at the top have the strongest global impact.

These are the Shap plots for each model:

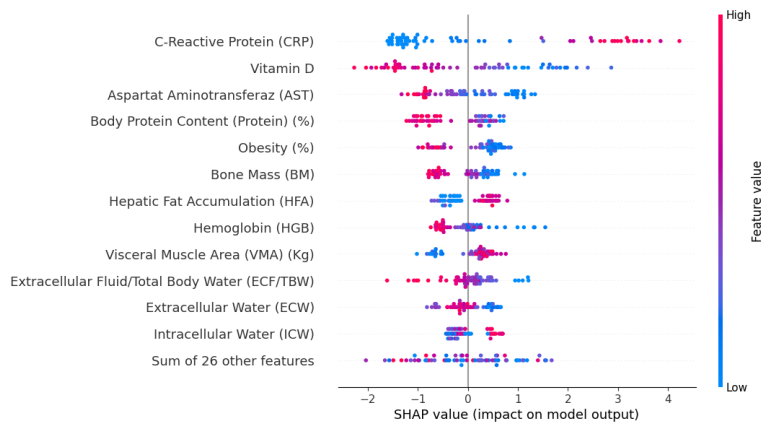
- SVM:



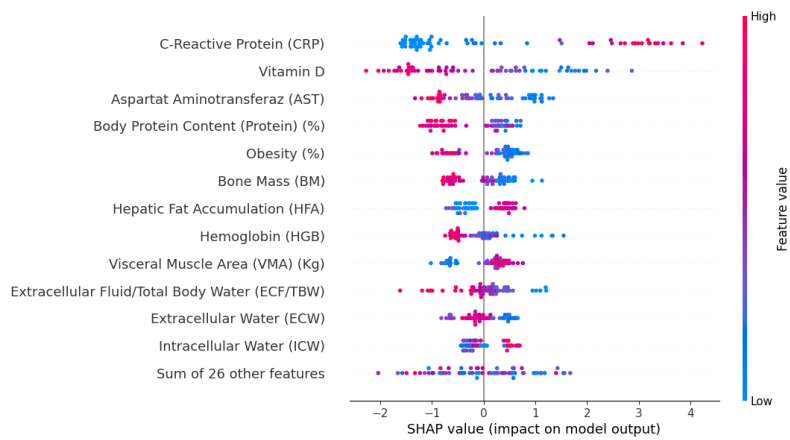
- Random Forest:



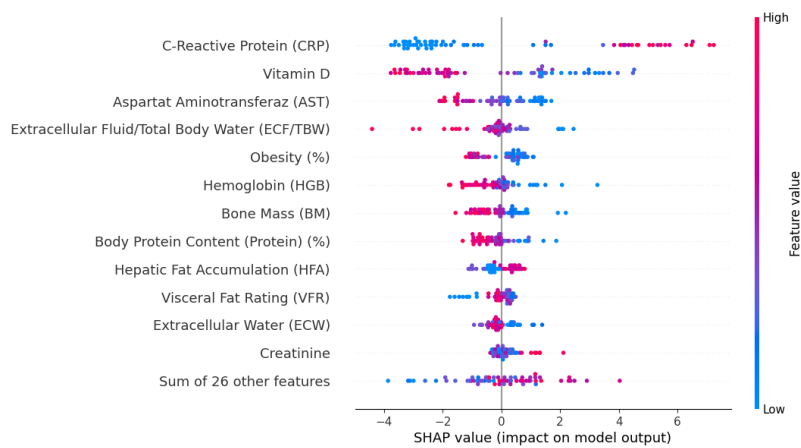
- XGBoost:



- Gradient Boosting:



- Decision Tree:



How Shap helps users:

1- Define the key biometric indicators:

The most consistent biometric indicators of all 5 models are:

- CRP was the number 1 feature for all model: higher CRP values move to the right and increase the risk (gallstone), lower CRP values have a neutral or slightly decreasing effect.
- Vitamin D was top 2 or 3 in all models: higher Vitamin D values move to the left and decrease the risk, lower Vitamin D values move to the right and increase the risk.

The user uses this information to always ensure checking CRP and Vitamin D first before Gallstone checking. The process will be cost effective, because the user will focus on checking only the important features. Moreover, it is good to build trust with patient by showing how each model makes the decision and which features affect the decision of the model.

2- Model transparency of decision-making:

Shap demonstrates how each feature contributed to a particular prediction, which helping users to comprehend the logical reasoning behind the model's results.

3- Trust and confidence in predictions:

The ability to visualize how each feature impacted the prediction builds trust among users, especially in sensitive applications such as healthcare because users require an understanding of why a particular decision was generated by the model.

4- Support for clinical decision making:

The use of Shap to provide explanations for model-based recommendations assists doctors to incorporate these recommendations into their diagnostic processes consistent with the way they would reason about a patient's condition.

5- Increase accountability of systems:

Shap provides a transparent auditing trail of how decisions are generated and is critical for ensuring compliance with both ethical and regulatory requirements.

6- Supports users' ability to interpret results from a systems level rather than treating the model as a Black Box.

7- Ensuring that the models' decisions are fair for everyone, including people in protected groups.

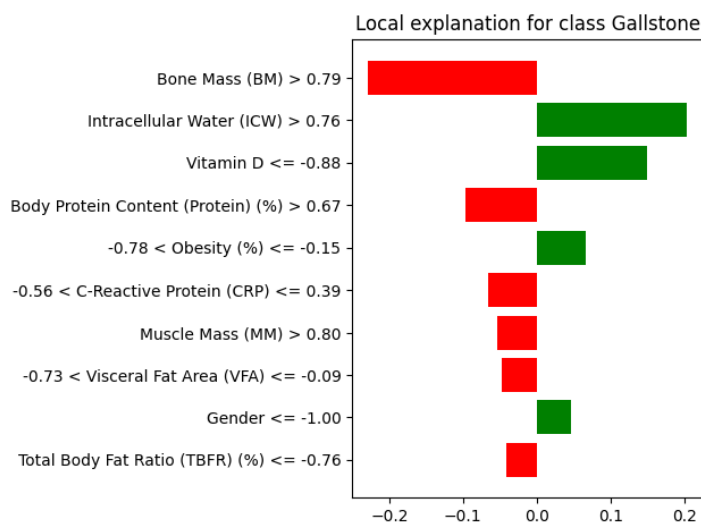
What Lime plot shows:

- **The Green bars:** represent features that have increased the prediction towards the positive class (Gallstone)
- **The Red bars:** represent features that have decreased the prediction.
- **Colour Coding:** Red [high feature values], Blue [low feature values]
- **The longer the green or red bar, the stronger the effect of that feature for that case.**

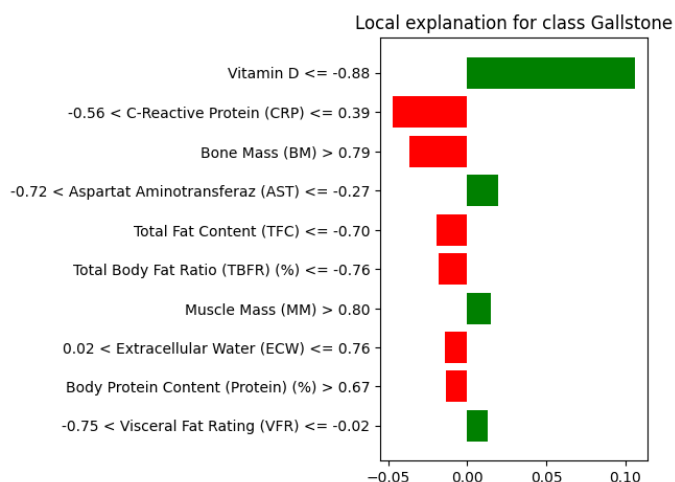
LIME provides a local explanation of the models using a focus on a single case and demonstrates how each feature affects that specific case's prediction. Features that appear at the top have a strong effect on that feature in that case. The LIME plot shows how different features affected the prediction for the case.

These are the Lime plots for each model:

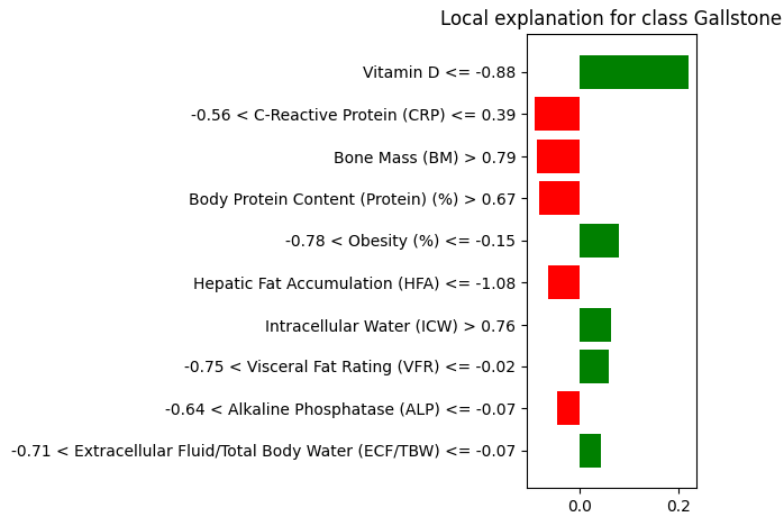
- SVM:



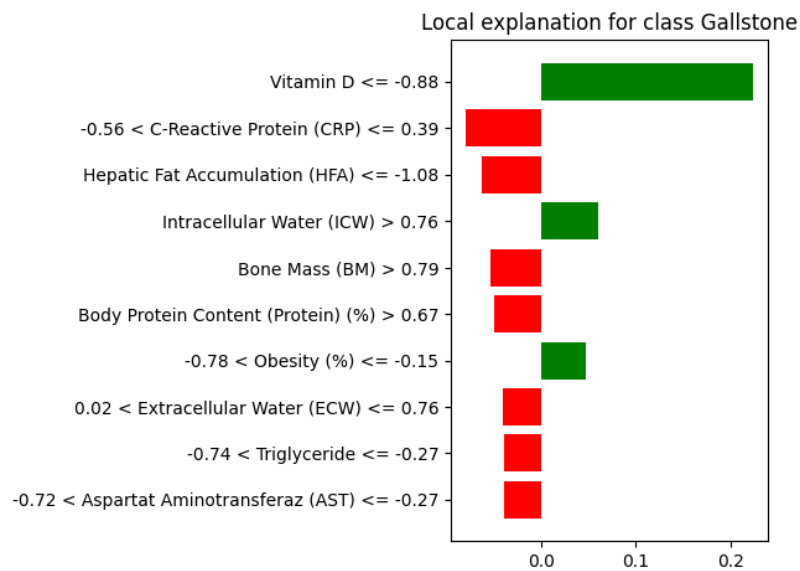
- Random Forest:



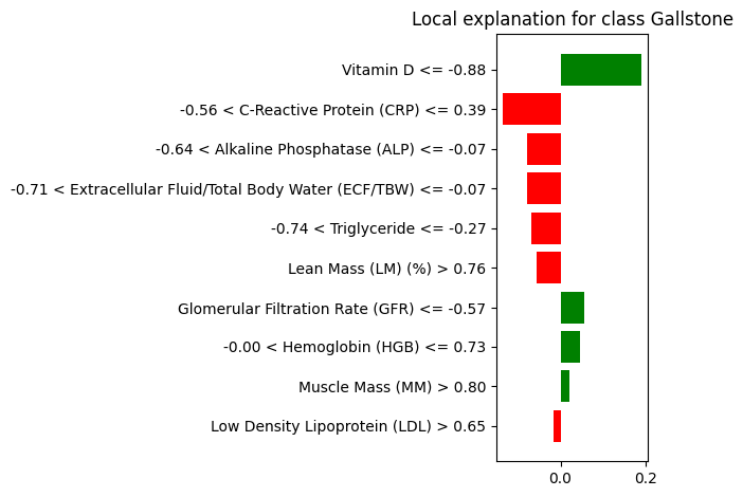
- **XGBoost:**



- **Gradient Boosting:**



- **Decision Tree:**



How Lime helps users:

1- Case-specific explanations:

It helps explain an individual patient's classification by the model into "Gallstone" or "No Gallstone".

2- Provides explanations for why a particular prediction was made for a specific individual.

3- Converts complex models' decisions into simple, user-understandable rules:

Because Lime is represented in bar chart with green colour for positive contributions and red colour for negative contribution.

4- Permits end-users to view which features were the most important contributors to the prediction for any user case.

5- Enables case-by-case interpretation:

which is crucial in sensitive areas like health care.

6- Serves to provide end-users with justification for their patient predictions when they make decisions

7- Non-technical users will find Lime's explanations useful:

to help them understand how different features contribute to the predicted outcomes for a specific case.

7 Critical evaluation and discussion (Q7)

1. How the method solves the problem:

The main objective of this project was to develop a machine learning model to diagnose gallstone disease using Gallstone Dataset that include patient demographics, body composition data obtained via bioimpedance, and laboratory test results, instead of using expensive imaging studies (ultrasound, CT, MRI).

SVM achieved the highest performance in terms of its test accuracy (87.5%), precision (89.65%), recall (83.87%), and F1-score (86.67%). These values represent strong performance on all metrics, indicating that the model accurately diagnoses patients with gallstones without needing to use invasive imaging modalities. XGBoost had the second-highest performance with an accuracy of 84.37%, and both precision and recall were relatively balanced at 83.87%. Gradient Boosting also showed moderate performance with an accuracy of 82.81%, and high precision of 85.71%, but low recall of 77.42%. Random Forest had moderate performance with an accuracy of 81.25%, whereas Decision Trees had the poorest performance with an accuracy of only 76.56%, and the lowest recall of 70.96%.

The models each successfully address some of the critical issues associated with the problem. The models provide a non-invasive method using only blood tests and body composition measurements, which are significantly cheaper than traditional imaging methods. The high accuracy of SVM shows that machine learning can be used to effectively screen for gallstone disease, and the system provides for early detection through identification of at-risk patients prior to symptom development. The well-balanced performance of SVM, including high precision to minimize false positive

diagnoses, and good recall to identify the majority of cases, make it appropriate for clinical use.

Although all three models had some successes, each of them is severely flawed. The model that produced the best results (SVM) missed about 16 percent of the actual patients who have gallstones (or 5 of 31 total patients), which is a critical clinical error since the treatment for gallstones can be a very serious condition if left untreated. In addition to the false negatives of 5-7 patients, the false positives of 3-6 patients in each model caused unnecessary anxiety and costs for additional testing. All four of the models were also extremely overfit to the data because their training accuracies all reached 100 percent while their test accuracies dropped off significantly. As such, none of the models demonstrated good generalizability to new patients.

2. Meeting users need and intend applications:

The design of the system allowed for an opportunity to serve multiple user groups; including but not limited to: Primary Care Physicians, Patients, and Healthcare Systems, each group having different needs.

Primary Care Physicians will need to have access to a simple and fast screening tool that can work with basic laboratory results. While the SVM model does achieve 87.5% accuracy in screening with standard labs (CRP and Vitamin D), it is somewhat difficult for some Primary Care settings to train and set up for use due to the need for feature scaling. While the Decision Tree model is the most interpretable of all models studied, with clear IF-THEN rules, it is unacceptable with regards to accuracy as it achieves 76.56% accuracy; therefore, it is not suitable for use in Primary Care despite being easy to use.

Doctors will need to assess the risk of disease comprehensively and to perform a thorough metabolic evaluation. So, using SVM model with the explainable AI Shap will help the doctor to see which metabolic factors are driving the risk in a global level.

Patients will also require understandable explanations of how individual risk factors influence their overall risk. The LIME plots provide patients with a visual representation of their personal risk factors through red and green bars, providing them with insight into which specific risk factors are contributing to their overall risk. Additionally, the system identifies actionable insights such as addressing inflammation or vitamin D deficiency.

Healthcare Systems will require cost reductions and effective methods for screening populations. The SHAP analysis indicates that if healthcare systems focus their screening efforts on CRP and Vitamin D screening, they will obtain the most critical information necessary to allocate resources efficiently. As a result, the 87.5%

accuracy achieved by SVM could potentially eliminate the necessity for ultrasound screenings, resulting in considerable cost savings.

3. Limitations:

- **Small Data set:**

First, the overall number of patients is only 319, and 255 were used to train the models while the remaining 64 were reserved for testing. Second, because the total amount of data is so limited, it severely impacts the degree of overfitting. The degree of overfitting is seen in four out of five models in the form of a large drop in test performance after achieving 100% training performance, ranging from 15-25%. As a result of this very small data set, the models are unable to learn patterns that are generally applicable, instead, they can memorize the individual training examples.

In addition, the fact that all the data was obtained from a single hospital in Turkey over the course of one year also limits the potential generalizability of the models across various demographics, geographic locations, and types of healthcare delivery systems.

- **Model Performance Limitation:**

Four out of five models demonstrated severe overfitting, with the largest differences in training and test accuracy being 11.68% to 18.75%, and cross-validation accuracy gaps of up to 24.71% for Gradient Boosting. Overfitting indicates that Random Forest, XGBoost, Gradient Boosting, and Decision Tree models are memorizing training data as opposed to finding generalizable patterns within the data. The inability to find generalizable patterns makes these models unreliable for use in clinical applications, even though they achieved perfect 100% training accuracy.

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4. Future Improvement:

- **Addressing Overfitting Through Data Expansion and Regularization:**

Addressing Overfitting Through Data Expansion and Regularization

The most critical improvement needed is expanding the dataset from 319 to at least 1,000 patients through multi-centre collaboration with 5-10 hospitals across different countries, which would provide the models with sufficient diverse examples to learn generalizable patterns rather than memorizing training data. Simultaneously, stronger regularization techniques should be applied including reducing Random Forest max_depth from 10 to 3-5, lowering XGBoost learning_rate from 0.2 to 0.01 with early stopping, and decreasing Gradient Boosting n_estimators from 200 to 50-100. Feature selection using SHAP importance scores should be implemented to build simpler models using only the top 10-15 most important features rather than all 38, which would reduce model complexity while maintaining predictive power.

- **External Validation and Prospective Clinical Trials:**

Partnerships with 3-5 hospitals in different countries should be established to obtain independent datasets for testing whether SVM accuracy remains above 85% on completely new populations with different genetic backgrounds and healthcare contexts. A prospective clinical trial should be designed to screen 500 new patients using model predictions and then confirm all cases with ultrasound imaging, tracking how many high-risk predictions actually develop gallstones over 1-2 years and measuring whether model-guided interventions such as vitamin D supplementation and anti-inflammatory treatment actually prevent disease formation.

- **Improving Recall:**

To address the critical issue of missing 16-29% of actual gallstone cases, the decision threshold should be lowered to classify more patients as at-risk, accepting some increase in false positives because from a medical perspective over-screening is safer than missing cases. Cost-sensitive learning should be implemented by assigning false negative penalties three times higher than false positive penalties, forcing models to prioritize catching all disease cases. An ensemble voting system combining SVM, XGBoost, and Gradient Boosting should be developed where patients are flagged if any two out of three models predict gallstone, which would likely increase recall above 90% while maintaining reasonable precision.

8 References

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